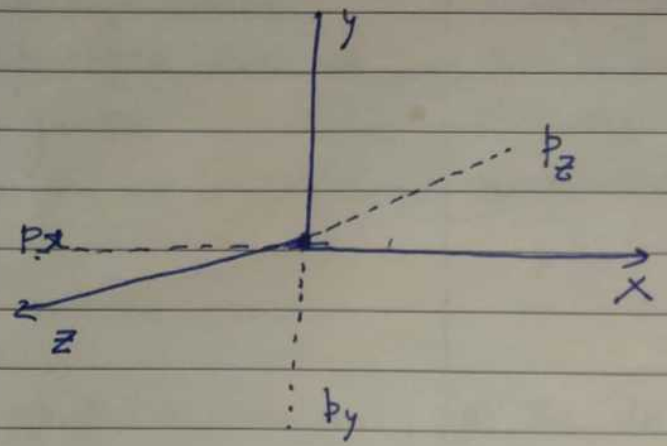
~~Statistical Mechanics~~

Quantum Statistics.



$$\delta V = dx\,dy\,dz \; \& \; dp_x\,dp_y\,dp_z$$

(cell vol)

(phase cell vol)

1) Maxwell Boltzman Statistic.: ~~Distinguishible identical part (gas)~~ (Mol.)
2) Bose Einstein Stat.
3) Fermi dirac stat.

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Maxwell Boltzman	Bose - Einstein	Fermi Dirac
1) Apply to system of distinguishable particles, such as gases	1) Apply to system of indistinguishible particles that do not obey Pauli's exclusion principle. eg: photons, phonons, liq He at low temp.	1) Apply to system of indistinguishible particles that obey Pauli's exclusion principle. eg free e^- in metal & stars (white dwarf)
2) The distribution law is $n_i = \frac{g_i}{e^{\alpha} e^{E_i/K_B T}}$	2) The distribution law is $n_i = \frac{g_i}{(e^{\alpha} e^{E_i/K_B T} - 1)}$	2) The distribution law is $n_i = \frac{g_i}{(e^{\alpha} e^{E_i/K_B T} + 1)}$
3) No restriction on the no. of particles in the given state	3) Same as Maxwell Boltzmann	3) only one particle in a given quantum state is allowed.
4) Vol of phase cell not fixed	4) Vol of phase cell is of the order $= h^3$	4) Same as Bose - Einstein
5) The behaviour of distribution func. $f(E_i)$ against E_i is exponential	5) For $E_i >> K_B T$, exponential for $E_i << K_B T$, lies above M.B.	5) For $E_i >> K_B T$, exponential where $E_i >> E_F$. If $E_F >> K_B T$ decrease abruptly near E_F
6) The energy of an M.B. system, such as in ideal monoatomic gas may be 0 at abs 0	6) The energy may be zero at abs. zero.	6) The energy at absolute zero, cannot be zero because all particles cannot come down to ground state due to Pauli Exclusion princip

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phase Space

- Any instaneous state of particle is represented by some points in an imaginary 6D space in which 6 co-ordinates are marked along the ~~axis~~ respective axis is called phase space or μ-space.

phase cell.

- phase space is divided into small 6D space whose sides are $\delta x, \delta y, \delta z$ or $\delta p_x, \delta p_y, \delta p_z$ are called phase cell.

- Vol (phase cell) $= \delta x\, \delta y\, \delta z\, \delta p_x\, \delta p_y\, \delta p_z = \delta V$

Micro State of system

→ At a pirticular instant be defined when we specify as to which particular cell each molecule of system belongs at that instant.

→ ~~WA~~

Macro state of System

→ It can be defined just by giving no. of molecule in each cell.

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Constraint

→ Energy of system constant.

→ Total particle in system is const.

Maxwell Boltzman Stats.

Let us consider a system of N distinguishible particle of a gas.

Let n_1 of them have energy E_2 & n_2 of them have energy E_2 so on and so forth.

thus the entire assembly of molecule can be divided into energy state with Energies $E_1, E_2, E_3, \cdots E_i$ having $n_1, n_2, \cdots n_i$ molecule

Cond^n

(a) Total no of molecule is constant.

$$N = n_1 + n_2 + n_3 + \cdots + n_i = \text{const.}$$

$$\delta N = \delta n_1 + \delta n_2 + \delta n_3 + \cdots + \delta n_i = 0$$

$$\sum_i \delta n_i = 0 \quad \text{——} (1)$$

(b) Total energy of molecule is const.

$$E = n_1 E_1 + n_2 E_2 + n_3 E_3 + \cdots + n_i E_i = \text{const.}$$

$$\delta E = \delta n_1 E_1 + \delta n_2 E_2 + \delta n_3 E_3 + \cdots + \delta n_i E_i = 0$$

$$\therefore \quad \sum_i E_i \delta n_i = 0 \quad \text{——} (2)$$

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(c) There are g_i cell with energy E_i. The total no. of ways in which n_i molecules have energy E_i is $(g_i)^{n_i}$

$(g_i)^{n_i}$

hence the total no. of ways N molecules can be distributed among various energy

$$\omega_1 = (g_1)^{n_1} \cdot (g_2)^{n_2} \cdots\cdots (g_i)^{n_i} \quad \text{—— (3)}$$

The no of ways in which grp. of particles can be choosen from total number of particle.

$$\omega_2 = \frac{\lfloor N}{\lfloor n_1 \lfloor n_2 \cdots \lfloor n_i} \quad \text{—— (4)}$$

The no. of distinguishible ways in which n molecules can be distributed among possible energy levels.

$$\omega = \omega_1 \omega_2 = \frac{\lfloor N}{\lfloor n_1 \lfloor n_2 \lfloor n_3 \cdots \lfloor n_i} (g_1)^{n_1} (g_2)^{n_2} \cdots (g_i)^{n_i} \quad \text{—— (5)}$$

This is the thermodynamic prob. of a system.

For most probable distr. ω is max^m subject to the restriction such that total no. of particle & total energy of system is constant.

Taking nat. log of Eqn (5), we get

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$$\ln \omega = \ln N! - \sum \ln n_i! + \sum n_i \ln g_i$$

Using Stirling thrm

$$\ln \omega = N \ln N - N - \sum n_i \ln n_i - n_i + \sum n_i \ln g_i$$

$$= N \ln N - \sum n_i \ln n_i + \sum n_i \ln g_i \quad \text{---} (6)$$

ω is max so change in $\omega = 0$

Stirling thrm

$$\ln n! = n \ln n - n$$

$$\therefore \delta \ln \omega_{max} = -\sum n_i \frac{1}{n_i} \delta n_i - \sum (\ln n_i)\, \delta n_i + \sum (\ln g_i)\, \delta n_i = 0$$

$$\Rightarrow \delta \ln \omega_{max} = -\sum (\ln n_i)\, \delta n_i + \sum (\ln g_i)\, \delta n_i = 0 \quad \text{---} (7)$$

Multiply Eqn (1) by $-\alpha$ & eqn (2) by $-\beta$ [α & β = boltzman const] & addit to Eqn 7.

$$\therefore \sum (-\ln n_i + \ln g_i - \alpha - \beta E_i)\, \delta n_i = 0$$

$$\Rightarrow -\ln n_i + \ln g_i - \alpha - \beta E_i = 0$$

$$\Rightarrow \ln n_i = \ln g_i - \alpha - \beta E_i$$

$$\Rightarrow n_i = e^{(\ln g_i - \alpha - \beta E_i)} = \frac{g_i}{e^{\alpha + \beta E_i}} = g_i\, e^{-(\alpha + \beta E_i)}$$

$$\therefore \boxed{n_i = g_i\, e^{-\alpha - \beta E_i}}$$

$\alpha = \frac{\mu}{K_B T}$ → chem pt. of gas , $\beta = \frac{1}{K_B T}$

$$\boxed{n_i = g_i / e^{\alpha + \frac{E_i}{K_B T}}}$$

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Application of Maxwell & Boltzman Law.

→ Molecules energy in an ideal gas.

Acc to Maxwell Boltzmann distribution law.

$$n_i = \frac{g_i}{e^{(\alpha + \frac{E_i}{k_B T})}} \quad \text{——— (1)}$$

Consider an ideal gas that contains N molecules. Consider a continuous distribution of molecular energies instead of the discrete set $E_1, E_2, E_3, \ldots$ then the eqn becomes.

$$n(E)\,dE = g(E)\, e^{-\alpha}\, e^{-(E/k_B T)}\, dE \quad \text{——— (2)}$$

$n(E)\,dE$: no. of molecule having Energies b/w E & E+dE

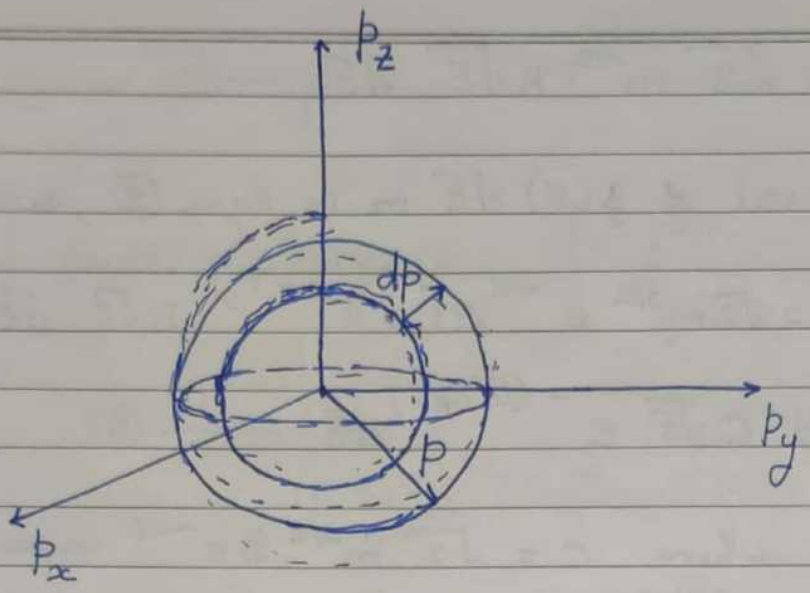
$g(E)\,dE$: Here, $g(E)dE$ represents: no. of states having energies b/w E & E+dE.

now, A molecule of energy E has a momentum p whose magnitude is

$$p = \sqrt{2mE} = \sqrt{p_x^2 + p_y^2 + p_z^2} \quad \text{——— (3)}$$

Each component p_x, p_y & p_z represent a different state of motion.

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Volume of the sphere $= \frac{4}{3}\pi p^3$

" " "

Spherical shell of thickness $dp = dV = 4\pi p^2 dp$

$\therefore$ $g(p)\,dp = Bp^2\,dp$ ——(4), where B is some constant

$g(p)\,dp$ → no. of states p having momentum ~~energies~~ b/w P & $P+dp$

$\therefore$ Each momentum of magnitude p corresponds to a single energy E, the number of energy states $g(E)\,dE$ and $E+dE$ is the same as no. of momentum states $g(p)\,dp$ b/w p & $p+dp$.

$\therefore$ $g(E)\,dE = Bp^2\,dp$

now, $p^2 = 2mE$ & $dp = ?$

$\therefore$ $2p\,dp = 2m\,dE \Rightarrow dp = \frac{2m\,dE}{2\sqrt{2mE}}$

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$$\therefore g(E)\,dE = \sqrt{2}\, m^{3/2} B \sqrt{E}\, dE \quad \text{—⑤}$$

Substituting val of $g(E)\,dE$ in Eqn. ②, we get

$$\therefore n(E)\,dE = \sqrt{2} m^{3/2} e^{-\alpha} e^{-(E/K_B T)} B\sqrt{E}\, dE$$

$$= C\sqrt{E}\, e^{-(E/K_B T)} dE \quad \text{——⑤}$$

where $C = \sqrt{2}\, m^{3/2} B e^{-\alpha}$

To find C we make use of the normalisation cond^n in Eqn ⑤

Total no of molecules, $N = \int_0^\infty n(E)\,dE$

$$= C \int_0^\infty \sqrt{E}\, e^{-(E/K_B T)} dE$$

$$= C \frac{1}{2(1/K_B T)} \sqrt{\frac{\pi}{(1/K_B T)}} = C \times \frac{\sqrt{\pi}}{2} \times (K_B T)^{3/2}$$

$$\Rightarrow C = \frac{2\pi N}{(\pi K_B T)^{3/2}} \quad \text{——⑥}$$

putting the Val of C from Eqn ⑥ to Eqn ⑤, we get

$$\therefore n(E)\,dE = \frac{2\pi N}{(\pi K_B T)^{3/2}} \sqrt{E}\, e^{-E/K_B T} dE \quad \text{——⑦}$$

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∴ Total Energy of the system is

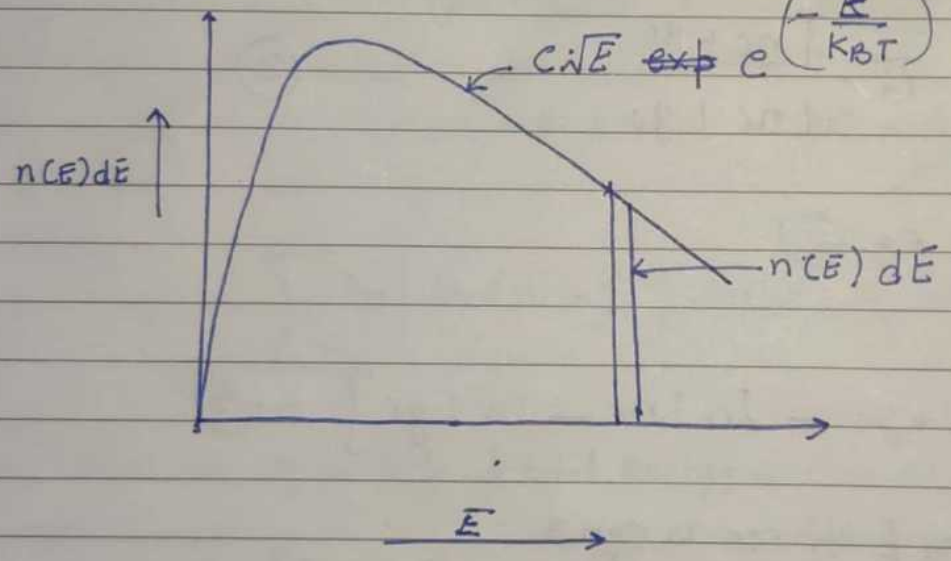
$$\bar{E} = \int_0^\infty \bar{E}\, n(\bar{E})\, d\bar{E} = \frac{2\pi N}{(\pi K_B T)^{3/2}} \int_0^\infty \bar{E}^{3/2}\, e^{-\bar{E}/K_B T}\, d\bar{E}$$

$$= \frac{2\pi N}{(\pi K_B T)^{3/2}} \times \frac{3}{4} (K_B T)^2 \sqrt{(\pi K_B T)}$$

$$= \frac{3}{2} N K_B T \quad \text{——} (8)$$

∴ Total Energy of system = $\frac{3}{2} N K_B T$

Avg " " gas molecule = $\frac{\bar{E}}{N} = \frac{3}{2} K_B T$



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Bose - Einstein Statistics

Consider an assembly of N bosons (identical & indistinguishible). No restriction is imposed to the number of particles that may occupy a given cell

Let us consider a box divided by g_i section by $(g_i - 1)$ partitions and indistinguishible particle to be seperated among section.

Total no. of distinguishible & distinct ways of arranging N particle in all variable energy states is given by

$$\omega = \pi \frac{\lfloor n_i + g_i - 1}{\lfloor n_i \, \lfloor g_i - 1} \quad \text{---} (1)$$

$$= \pi \frac{\lfloor n_i + g_i}{\lfloor n_i \, \lfloor g_i} \quad \text{---} (2)$$

Taking ln of eq. (2)

$$\ln \omega = \sum \left[\ln \lfloor n_i + g_i - \ln \lfloor n_i - \ln \lfloor g_i \right] \quad \text{---} (3)$$

Applying stirling approx in eqn 3

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$$\ln w = \sum (n_i + g_i)\ln(n_i + g_i) - (n_i + g_i) - n_i \ln n_i + n_i - g_i \ln g_i + g_i$$

$$\Rightarrow \ln w = \sum \left[(n_i + g_i)\ln(n_i + g_i) - n_i \ln n_i - g_i \ln g_i \right] \quad -(4)$$

Here,
since g_i is not variable & n_i varies continuously

For most probable distribution.

$$\delta \ln(w)_{max} = 0$$

$$\delta \ln w_{max} = \sum (n_i + g_i) \times \frac{1}{(n_i + g_i)} \delta n_i + \ln(n_i + g_i)\delta n_i - \delta n_i \ln n_i - n_i \times \frac{1}{n_i} \delta n_i$$

$$= \sum \left[\ln(n_i + g_i)\,\delta n_i - \delta n_i \ln n_i \right] = 0 \quad -(6)$$

Total no. of particle & total energy is const

$$\sum \delta n_i = 0 \quad -(7)$$

$$\sum \delta n_i E_i = 0 \quad -(8)$$

multiply eqn (7) by $-\alpha$, (8) by $-\beta$ & adding in eqn (6)

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$$\sum \left[\ln(n_i + g_i) - \ln n_i - \alpha - \beta E_i \right] \delta n_i = 0$$

$$\Rightarrow \ln(n_i + g_i) - \ln n_i - \alpha - \beta E_i = 0$$ where

$$\cdot \alpha = \frac{-\mu}{k_B T}$$

$$\beta = -\frac{1}{k_B T}$$

$$\Rightarrow \ln\left(\frac{n_i + g_i}{n_i}\right) = \alpha + \beta E_i$$

$$\Rightarrow \boxed{n_i = \frac{g_i}{(e^{\alpha + \beta E_i} - 1)}}$$

$$\Rightarrow n_i = \frac{g_i}{(e^{\alpha + \beta E_i / k_B T} - 1)}$$

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Fermi - Dirae Statistics

Consider N fermions with total Energy E suppose that n_i particle occupy the energy level E_1 & n_2 particle occupy the energy level E_2 & so on.

Let us find out the total no of ways in which n_i particles can be distributed in g_i cells having energy E_i.

$$W = \prod \frac{\lfloor g_i}{\lfloor n_i \; \lfloor g_i - n_i} \quad \text{——— (1)}$$

Taking natural log of Eqn (1)

$$\ln W = \sum \ln \lfloor g_i - \ln \lfloor n_i - \ln \lfloor g_i - n_i \quad \text{——— (2)}$$

Applying Stirling approx in eqn (2)

$$\ln W = \sum g_i \ln g_i - g_i - n_i \ln n_i + n_i - (g_i - n_i) \ln (g_i - n_i) + g_i - n_i$$

$$= \sum g_i \ln g_i - n_i \ln n_i - (g_i - n_i) \ln (g_i - n_i)$$

For most probable distr.

$$\delta \ln W_{max} = 0 = \sum 0 - n_i \times \frac{1}{n_i} \delta n_i - \ln n_i \, \delta n_i + (g_i - n_i) \times \frac{1}{(g_i - n_i)} \delta n_i + \ln (g_i - n_i) \, \delta n_i$$

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$$\Rightarrow \delta(\ln W_{max}) = \sum \left[-\ln n_i + \ln(g_i - n_i)\right] \delta n_i = 0 \quad \text{---(3)}$$

(2) $\sum \delta n_i = 0$ ---(4)

(3) $\sum \delta n_i E_i = 0$ ---(5)

$$-\alpha = \frac{E_F}{K_B T}$$

$$-\beta = \frac{1}{K_B T}$$

multiply (4) & (5) with $-\alpha$ & β resp and add in eqn (3)

$$\sum \left[-\ln n_i + \ln(g_i - n_i) - \alpha + \beta E_i\right] \delta n_i = 0$$

$$\Rightarrow \ln(g_i - n_i) - \ln n_i = \alpha + \beta E_i$$

$$\Rightarrow \frac{g_i - n_i}{n_i} = e^{\alpha + \beta E_i}$$

$$\Rightarrow n_i \left(e^{\alpha + \beta E_i} + 1\right) = g_i$$

$$\Rightarrow n_i = \frac{g_i}{e^{\alpha + \beta E_i} + 1} = \frac{g_i}{\left(e^{\alpha + \frac{E_i}{K_B T}} + 1\right)} = \frac{g_i}{\left(e^{\frac{(E_i - E_F)}{K_B T}} + 1\right)}$$

At abs 0 the max energy of free e^- is called fermi energy

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Application of Fermi Dirac Stats.

Till $E_F = \frac{h^2}{2M}\left(\frac{3N}{8\pi V}\right)^{2/3}$

(a) Fermi Energy

Consider a system of N fermions enclosed in a volume V. We know that because of Pauli's principle, only one fermion can be accommodated in a given state. It is also known to us that the highest energy possessed by a fermion at T = 0K is called Fermi Energy, E_F. Let us,

We know that the number of quantum states of a particle with momentum in the interval p & $p+dp$ is $\frac{4\pi V}{h^3} p^2 dp$. we multiply this number by $(2S+1)$. For e^-, $S = \frac{1}{2}$ so that the required no. of states is $\frac{8\pi V}{h^3} p^2 dp$.

Denoting the highest momentum p_F, we have.

$$N = \frac{8\pi V}{h^3}\int_0^{P_F} p^2\,dp = \frac{8\pi V}{h^3}\,\frac{p^3}{3}$$

This yeilds an expr. for fermi momentum, p_F.

$$p_F = \left(\frac{3N}{8\pi V}\right)^{1/3} h$$

& fermi Energy

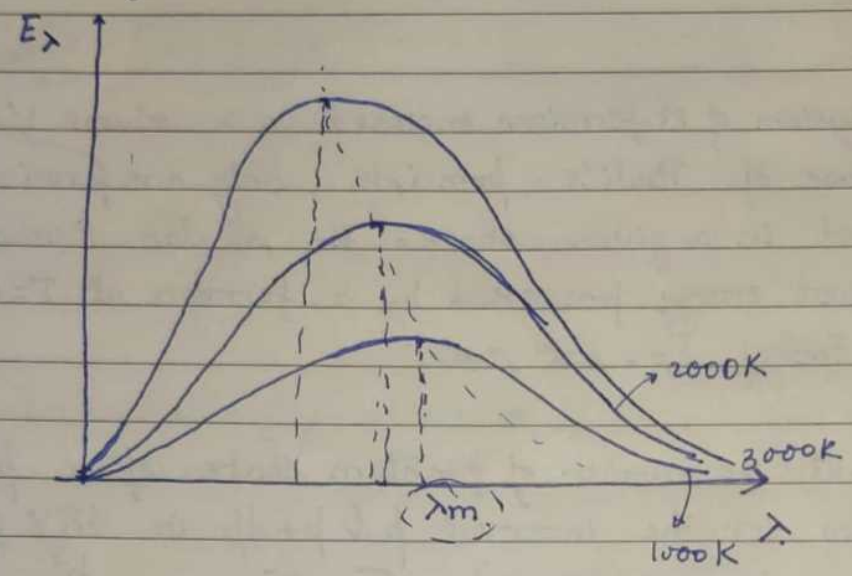
$$E_F = \frac{p_F^2}{2m} = \frac{h^2}{2m}\left(\frac{3N}{8\pi V}\right)^{2/3}$$

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B

Application of Base Einstein.

i) Black body radiation



$\lambda m T = cost$

$E = \sigma T^4$ Stephen Laws

$E_\lambda d\lambda \propto \frac{1}{\lambda^5}$ for lower λ

$E_\lambda d\lambda \propto \frac{1}{\lambda^4}$ for higher λ

When a black body is heated it emits radiations of all types of wavelength, these radiations are indipendent of nature of substance such heat radiations ie in a uniform temp. enclosure are known as blackbody radiation.

eg: Pt. black.

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Variation of Intensity of Black Body Radiation.

- When a blackbody is heated at diff. temp. it emits radiations of all types of wavelength. The origin of radiation from heated body is the rapid vibrating particles. Known as oscillators.

- Acc maxwell em theory these ascillator ~~radiate~~ radiate energy in the form of EM waves. That can be shown by the graph of variation of Intensity of radiation. with the wavelength at diff temp.

Following Conclusion can be drawn:

- At a given temperature the energy is not uniformly distributed. In the ~~variation~~ radiation spectrum of the hot body.

- At ~~regular~~ given temperature the intensity of Rad. inc. with inc. in wavelength. At a particular λ its val. is max. Beyond that the intensity of heat radiation decreases.

- ~~Why~~ Wein's disp. law : $\lambda_m T = \text{const.}$
  With inc. in temp. λ_m decr. such that

$$\lambda_m T = \text{const} = 0.2898 \text{ cm K}$$

Where λ_m is the wavelength at which max^m emission of energy takes place. This is called Wein's disp law.

For all wavelengths an increase in temp causes an.

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- increase in energy emission.

- Stephen's law : The area under each curve repr. the total energy emitted by the body at ~~pir~~ particular temp for the range of wavelength considered. This area inc. with inc in temp. The Area/energy $\propto T^4$. This is called Stephen's law.

$$E \propto T^4$$

Limitation of Classical Mechanic for Black body radiation

Rayleigh - Jean's law agrees with the expr. result at higher wave length. but does not fit the curve in the lower wavelength region. While the Wein's Energy distr. agrees with the expr. curve for shorter wave length region but failed for higher wavelength. The freq distribution of black body could not be explained acc. to classical concepts so ~~plank's~~ plank's radiation formula on the basis of quantum theory was reqd.

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Plank's Radiation Law (Plank's hypothesis)

- Plank derieved the theoritical expression for energy distr. in black body on the basis of quantum theory of heat radiation by making following assumption.

→ Black body chamber is filled up not only with radiation but also with Simple harmonic oscillator or resonators of molecular dimension called as oscillator. which can vibrate with all possible frequencies, but retains with one degree of freedom.

→ Classical principle of Equipartition of energy is not applicable to black body oscillator. instead it is assumed that the oscillator of a black body can have discreate energy = integral multiple of $h\nu$ $\Rightarrow E = nh\nu$

→ Oscillator cannot radiate or absorb energy continiously however the oscillator can exchange energy with its surrounding in discreate units called quanta.

Inorder to derive plank radiation law first derive no. of resonator / vol. lying the freq range ν & $\nu + d\nu$ & avg energy of plank's resonator.

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Planck's radiation law

Let N be the total no of oscillators & E be the total Energy. then energy/oscillator can be given as $\frac{E}{N}$.

$$\bar{\varepsilon} = \frac{E}{N} \quad \text{---(1)}$$

$N_0, N_1, N_2, \ldots N_r \ldots$ are the no of oscillators, having Energy $0, \varepsilon, 2\varepsilon, \ldots r\varepsilon \ldots$ etc respectively. Then total no of particle.

$$N = N_0 + N_1 + N_2 + N_3 + \ldots + N_r + \ldots \quad \text{---(2)}$$

$$E = 0 + N_1\varepsilon + 2\varepsilon N_2 + 3\varepsilon N_3 + \ldots + N_r r\varepsilon + \ldots \quad \text{---(3)}$$

Acc to B maxwell distr. eqn the no. of particle having energy $r\varepsilon$ is

$$N_r = N_0 e^{-r\varepsilon/KT} \quad \text{---(4)}$$

$$N = N_0 e^{0} + N_0 e^{-\varepsilon/KT} + N_0 e^{-2\varepsilon/KT} + \ldots N_0 e^{-r\varepsilon/KT} + \ldots$$

$$= N_0 \left[1 + e^{-\varepsilon/KT} + e^{2\varepsilon/KT} + \ldots\right]$$

$$= N_0 \left(\frac{1}{1 - e^{-\varepsilon/KT}}\right) \quad \text{---(5)}$$

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$$E = 0 + \frac{E}{N} + \frac{2E}{N} + \cdots$$

$$E = N_1\varepsilon + 2\varepsilon N_2 + 3N_3\varepsilon + \cdots$$

$$= \varepsilon\left[N_1 + 2N_2 + 3N_3 + \cdots + N_r r + \cdots\right]$$

$$= \varepsilon\left[N_0 e^{-\varepsilon/kT} + 2N_0 e^{-2\varepsilon/kT} + \cdots + rN_0 e^{-r\varepsilon/kT} + \cdots\right]$$

$$= N_0\varepsilon e^{-\varepsilon/kT}\left[1 + 2e^{-\varepsilon/kT} + 3e^{-2\varepsilon/kT} + \cdots + re^{-(r-1)\varepsilon/kT} + \cdots\right]$$

$$= \frac{N_0\varepsilon e^{-\varepsilon/kT}}{\left(1 - e^{-\varepsilon/kT}\right)^2} \quad \text{---} (6)$$

$$\therefore \bar{\varepsilon} = \frac{N_0\varepsilon e^{-\varepsilon/kT}}{\left(1 - e^{-\varepsilon/kT}\right)^2} = \frac{\varepsilon \cdot e^{-\varepsilon/kT}}{1 - e^{-\varepsilon/kT}}$$

$$N_0\left(\frac{1}{1 - e^{-\varepsilon/kT}}\right)$$

DOMS

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ν

$$\Rightarrow \bar{\varepsilon} = \frac{\varepsilon}{(e^{\varepsilon/kT} - 1)}$$

$$\Rightarrow \bar{\varepsilon} = \frac{h\nu}{(e^{h\nu/kT} - 1)}$$

The number of oscillators/vol in freq range ν & $\nu + d\nu$

$$N = \frac{8\pi\nu^2}{c^3} d\nu$$

avg energy of oscillator on multiplying by the no of oscillator/vol in the freq range of ν & $\nu + d\nu$ is

$$E_\nu d\nu = \frac{8\pi\nu^2}{c^3} d\nu \times \frac{h\nu}{e^{h\nu/kT} - 1}$$

$$\Rightarrow E_\nu d\nu = \frac{8\pi h\nu^3}{c^3} \left(\frac{1}{e^{h\nu/kT} - 1}\right) d\nu \quad \text{— (9)}$$

DOMS

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$$\nu = \frac{c}{\lambda}$$

$$|d\nu| = \left| -\frac{c}{\lambda^2} d\lambda \right|,$$

$$\Rightarrow E_\lambda \left(\frac{c}{\lambda^2}\right) d\lambda = \frac{8\pi h^3}{c^3} \left(\frac{1}{e^{hc/\lambda^2 kT} - 1}\right) \frac{c}{\lambda^2} d\lambda$$

$$\Rightarrow \lambda E_\lambda \propto \frac{8\pi h^3}{c^3}$$

$$\Rightarrow E_\lambda \, d\lambda = \frac{8\pi hc}{\lambda^5} \left(\frac{1}{e^{hc/\lambda kT} - 1}\right) d\lambda \quad \text{— (10)}$$

Reduction of Wein's law & Rayleigh's law from plank radiation law

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Crystal Structure

Dying of a Star

Stages of a Star

(i) Nebula: A star forms with from a massive clouds of dust and gas in space, also called nebula. Nebulae are mostly ~~comprise~~ composed of hydrogen. Gravity begins to pull the gases together

(ii) Proto Star: As the mass falls together it gets hot. A star is formed when it is hot enough for the hydrogen nuclei to fuse together to make helium. The fusion process releases energy, which keeps the core of the star hot.

(iii) Main Sequence Star: During this stable phase in the life of a star, the force of gravity holding the star together is balanced by higher pressure due to the high temperatures. The Sun is at this stable phase in its life

(iv) Red giant star: When all the hydrogen has been used up in the fusion process, larger nuclei begin to form and the star may be expand to become a red giant.

DOMS

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(V) Super nova: A larger mass with mass (8-15x mass of sun) will do nuclear reaction ~~we~~ until it explodes as a super nova. Then it will form ~~either a dwarf star or~~ neutron star.

(VI) Hyper nova: Stars with mass (30x mass of sun) will cause hypernova by getting ~~pr~~ compressed due to its own gravity forming black hole

(VII) black hole: It is a cosmic body of extremely intense gravity from which even light cannot escape.

DOMS

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